

## 1 CODE EQUATIONS

SCOPE:  $W \leq 20'-0"$ ; simply supported prismatic wood rafter or joist spans ridge to eave.

ASCE 7-16 Eq. 7.3-1:

$$p_f = 0.7C_e C_t I_s p_g \quad (7.3-1)$$

$p_f$  flat roof snow load, psf  
 $C_e$  exposure factor, Table 7.3-1  
 $C_t$  thermal factor, Table 7.3-2  
 $I_s$  snow importance factor, Table 1.5-2  
 $p_g$  ground snow load, psf

ASCE 7-16 Eq. 7.4-1:

$$p_s = C_s p_f \quad (7.4-1)$$

$p_s$  balanced sloped roof snow load, psf  
 $C_s$  roof slope factor, Section 7.4

ASCE 7-16 Section 7.6.1 and Figure 7.6-1, narrow roof rafter system:

$$\text{leeward pressure} = I_s p_g, \quad \text{windward pressure} = 0$$

$W$  horizontal eave-to-ridge distance  
 Narrow roof rafter system  $W \leq 20'-0"$ ; simply supported prismatic member spans ridge to eave  
 Code location Section 7.6.1 prose; load pattern shown in Figure 7.6-1; no equation number

Governing uniform roof snow pressure for this note's scope:

$$q_u = \begin{cases} \max(p_s, I_s p_g) & 0.5:12 \leq \text{roof slope} \leq 7:12 \\ p_s & \text{roof slope} < 0.5:12 \text{ or roof slope} > 7:12 \end{cases}$$

$q_u$  governing unfactored uniform roof snow pressure, psf  
 Guard If the scope condition above is not met, use the general pattern in Section 7.6.1 and Figure 7.6-1

## 2 TWO-PATH CHECK

